

BOB-094 “ Tracker-fetch stress + chaos automation evidence

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Scope

tests/stress/test_tracker_fetch_stress_chaos.py “ Â§11.4.85 stress + chaos coverage for SearchOrchestrator.fetch_torrent (the outbound HTTP + response classification path in download-proxy/src/merge_service/search.py).

Scenarios delivered (8 + 1 meta = 9 tests, all GREEN)

STRESS

#	Test	Category
1	test_stress_sustained_100_sequential_fetches	sustained-load “ p50/p95/p99 over 100 iters
2	test_stress_concurrent_10_parallel_no_fd_leak	concurrent-contention “ no deadlock, no fd leak
3	test_boundary_empty_url_returns_none_or_error	boundary “ empty URL
4	test_boundary_max_length_url_survives	boundary “ 2000-char URL
5	test_boundary_off_by_one_content_type_edges	boundary “ CT classifier edges

CHAOS

#	Test	Category
6	test_chaos_network_drop_20pct_categorised_as_network	network-fault “ Â§11.4.69 feature_class:
7	test_chaos_midflight_kill_clean_degradation	process-death “ HTTP server kill fetch
8	test_chaos_input_corruption_malformed_response_quarantined	input-corruption “ malformed body quarantined

Test

9 test_section_114_85_category_map

Category

meta " mecha
coverage-claim
assertion

§11.4.115 RED capture (production-mutation-driven)

Scenarios with captured RED:

test_chaos_input_corruption_malformed_response_quarantined AND
test_boundary_off_by_one_content_type_edges.

Applied always-accept mutation to the CT/prefix classifier at download-proxy/src/merge_service/search.py:1893-1898 (replaced the boolean guard with if True: so a malformed HTML body would be returned to the caller). Both tests FAILED with user-observable assertion errors proving the classifier bypass was detected " see evidence/red_capture_against_mutated_classifier.txt.

Production tree restored (git diff --stat empty); re-ran the full suite " 9/9 GREEN " see evidence/green_capture_after_restore.txt.

Evidence artefacts

evidence/*.json " one per scenario, written by the tests themselves via the Python analogue of ab_pass_with_evidence (path exists + non-empty + parses). Each carries section=11.4.85, feature_class per §11.4.69 taxonomy, and user-observable metrics (latency percentiles, byte-equality, FD deltas, error-category counts, quarantine counts).

Discipline

- §11.4.161 rootless " no root, hermetic (in-process aiohttp stub + ephemeral loopback HTTP server for the mid-flight-kill scenario)
- §11.4.5 captured evidence " per-iteration latency histogram, categorised error counts, state deltas
- §11.4.14 chaos cleanup " atexit + finally teardown for the real HTTP server (no bound port left behind)
- §11.4.69 taxonomy " feature_class recorded per scenario
- §11.4.201 no false-null " FD-leak check honestly reports unmeasurable_on_this_host when /proc/<pid>/fd is unavailable
- §11.4.113 " no force-push in this delivery